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Early Grooming Detection Using Language and Emotion Analysis

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ABSTRACT

Online grooming represents a significant threat to children's safety on the Internet, as predators increasingly exploit online communication platforms to establish trust, manipulate victims and facilitate harmful interactions. The gradual and deceptive nature of grooming makes its early detection particularly challenging, requiring systems capable of identifying subtle behavioral and linguistic patterns before explicit predatory intent becomes apparent.

This thesis presents a hybrid approach for the early detection of online grooming conversations by combining semantic language representations with emotion-based features. The proposed method integrates contextual embeddings generated by transformer-based language models with fine-grained emotional representations extracted using the GoEmotions framework. By fusing these complementary sources of information, the system aims to capture both linguistic patterns and the emotional characteristics that emerge throughout grooming intersections.

To evaluate the proposed approach, the PAN12 Sexual Predator Identification dataset was preprocessed into progressive conversation windows, enabling the simulation of realistic early detection scenarios where only a portion of a conversation is available. The fusion architecture combines transformer-derived semantic features with emotion probability vectors through feature concatenation, allowing the classifier to leverage both contextual understanding and emotional profiling when making predictions.

Additionally, an interface based application is provided to further support the usefulness of the proposed model. This also provides information on how well the model would perform in a real life scenario with unseen conversations.

The results indicate that combining semantic and emotional representations constitutes an effective strategy for early online grooming detection and highlights the potential of hybrid deep learning models for supporting child protection in online environments.

Contents

Chapter 1

Introduction

1.1 Motivation and Objectives

In an era where social media platforms are the main medium for interactions between people all around the world and minors nowadays have access to it from a very young age, a lot of malicious acts have adapted to the online space. The online communication platforms have become environments where ill intended individuals can exploit vulnerable users, particularly children and adolescents. Among the various online threats, grooming represents a serious concern due to its deceptive nature and potentially sever consequences for victims.

Online grooming is typically a gradual process in which a predator attempts to establish trust, manipulate emotions and develop a relationship with a child before introducing harmful or exploitative intensions. Due to the fact that grooming does not imply physical harm and only produces serious psychological traumas is often unrecognized by parental figures or law enforcements. Grooming behaviors often emerge progressively throughout a conversation and detecting them only after the interaction has fully developed poses a grater threat to the child. Consequently, there is a growing need for automated systems capable of identifying suspicious behavior as early as possible.

Recent advances in Natural Language Processing and Deep Learning have enabled the development of models capable of capturing complex linguistic patterns within conversational data. At the same time, psychological studies have shown that grooming interactions contain distinctive emotional characteristics that may provide valuable information for detection.

The main objective of this thesis is to investigate whether combining semantic language representations with emotional information can provide a solution for the early detection of online grooming conversations task. To achieve this objective, transformer-based language models and emotion-based representations are integrated into a hybrid detection framework and evaluated on the PAN12 Sexual Predator Identification dataset using an early detection setting.

1.2 Main Contributions

The main contributions of this thesis are the following:

- The design and implementation of a preprocessing pipeline that transforms the PAN12 Sexual Predator Identification dataset into progressive conversation windows suitable for early detection experiments.
- The implementation and evaluation of several baseline and transformer-based approaches for online grooming detection, including TF-IDF with Logistic Regression, DistilBERT and RoBERTa.
- The integration of emotional representations generated by the GoEmotions framework in order to capture psychological and behavioral signals present in grooming conversations.
- The proposal of two hybrid feature fusion architectures that combine transformer-derived semantic embeddings with emotional-based features.
- A comprehensive experimental evaluation of all models at multiple stages of conversation progression, allowing the assessment of early detection capabilities.
- The development and deployment of a web-based application that integrates the best-performing fusion model and enables the analysis of previously unseen conversations through an intuitive user interface.

Chapter 2

Literature Review and Related Work

In recent years, many have taken on the task of detecting online grooming through various methods. Starting from the Sexual Predator Identification task of 2012, people have tried using different tools from traditional machine learning approaches or shifting towards deep learning techniques focusing on text classification. Naturally, some of them decided to improve their work by first understanding the psychological foundations of grooming. Through all of this, a good foundation for the task of identifying predators and online grooming detection was established. The following section will describe the analysis of the PAN12 dataset and some of the most effective and commonly used solutions currently available.

2.1 Exploratory Data Analysis of the PAN12 Sexual Predator Identification Dataset

PAN is a series of scientific events and shared tasks on digital text forensics and stylometry [ICJ+12]. In 2012, they held the International Sexual Predator Identification Competition, in which the PAN12 corpus was released and is one of the most widely used datasets for research in online grooming detection. It contains a large collection of chat conversations collected from multiple sources, including general Internet Relay Chat (IRC) logs, Omegle chats and even predatory interactions obtained from the Perverted Justice archive [ICJ+12]. The dataset is found in XML format, each entry containing the conversation id, where we can find message line, author, timestamp and text, as it can be seen in Figure 2.1.

The corpus comprises two major subsets: a training set and a test set. Their specification being shown in Table 2.1. Across the full corpus (train + test), the dataset includes approximately 222,055 conversations, with a small fraction exhibiting predatory behavior, reflecting the extreme class imbalance typical in online grooming detection research.

```
<conversations>
  ...
  <conversation id="0042762e26ed295a8576806f5548cad9">
    <message line="3">
      <author>f069dbec9ab3e090972d432db279e3eb</author>
      <time>03:20</time>
      <text>whats up?</text>
    </message>
    <message line="4">
      <author>f069dbec9ab3e090972d432db279e3eb</author>
      <time>03:21</time>
      <text>how u doing?</text>
    </message>
    ...
    <message line="10">
      <author>f069dbec9ab3e090972d432db279e3eb</author>
      <time>04:00</time>
      <text>sse you llater?</text>
    </message>
  </conversation>
  ...
  <conversation id="0209b0a30c8eced86863631ada73a530">
    <message line="3">
      <author>0042762e26ed295a8576806f5548cad9</author>
      <time>01:17</time>
      <text>and that i dont touch u</text>
    </message>
  </conversation>
  ...
<conversations>
```

Figure 2.1: Example of data form PAN12 (Source:[ICJ+12])

	Conversations	Users	Sexual Predators
Training	66.927	97.689	142
Testing	155.128	218.702	254

Table 2.1: PAN12 dataset entries

In terms of messages, the data set also includes cybersex between consenting adults among non-predatory conversations, making distinguishing grooming chats especially difficult [MV21]. They divided the chats into segments whenever a conversation was interrupted for more than 25 minutes and filtered all segments with more than 150 messages. This results in a total of 222k segments, of which 2.58% are grooming chats, through which the organizers try to mimic the distribution of grooming in actual online conversations [MV21].

Unfortunately, the PAN12 corpus has some limitations. All grooming conversations are essentially not real, being from decoy operations and not with actual victims, but, on the other hand, the non-grooming conversations are real. One of the biggest problems is that the dataset is separated into relatively short, unordered segments, thus completely blurring the true timeline of a chat [MV21].

Overall, the PAN12 dataset has its challenges in both traditional machine learning methods and deep learning models. It requires more attention than usual through different strategies, using training techniques that can handle imbalanced data, such as Class Weight or Focal Loss, careful feature design, or resampling techniques.

2.2 Traditional Machine Learning Approaches

Early research in the field focused on creating efficient architectures to handle large volumes of data while capturing the structural characteristics of predatory interactions. One of the pioneering methodologies was developed by Villatoro-Tello et al. (2012), who introduced a two-stage architecture designed to reduce computational complexity [VTJGE+12]. This system first identifies "Suspicious Conversations" (SCI) before performing a granular "Victim From Predator" (VFP) disclosure to isolate the offender [VTJGE+12]. By utilizing Support Vector Machines (SVM) and Neural Networks alongside Bag-of-Words (BoW) representations, this approach achieved a notable precision of 0.9804, as shown in Table 2.2, demonstrating that tiered filtering is highly effective for large-scale monitoring [VTJGE+12].

Run	Recall	Precision	F-measure	$F_{0.5}$
Baseline	0.4055	0.9537	0.5691	0.7507
SCI(NN-B) and VFP(NN-TF-IDF)	0.7874	0.9479	0.8602	0.9107
SCI(NN-B) and VFP(NN-B)	0.7874	0.9804	0.8734	0.9346

Table 2.2: Official evaluation results from the submitted runs (Source:[VTJGE+12])

Escalante et al. (2013) recognized that grooming is not a static event, but rather a sequence of steps that happen over time. With this idea in mind, they went beyond global classification by proposing a chain classifier approach that segments conversations into three distinct phases: gaining access, developing a deceptive relationship and pursuing a sexual affair [EVTJ+13]. Their system uses the classifier chain to make predictions from earlier stages that will serve as features for subsequent ones [EVTJ+13]. This captures the temporal dependencies of the predator's "course of conduct," with results showing that segment-specific models significantly outperform global classifiers, particularly in the initial stages of grooming, Table 2.3.

Setting	Precision	Recall	F_1 Measure
Global	95.14%	49.91%	65.42%
Segment 1	96.16%	59.20%	73.23%
Segment 2	96.25%	48.82%	64.72%
Segment 3	93.43%	51.87%	66.68%

Table 2.3: Performance of global and local classifiers (Source:[EVTJ+13])

2.3 Deep Learning Approaches

As a natural course of action, Ebrahimi et al. (2016) decided to shift to a deep learning approach. They implemented a Convolutional Neural Network (CNN) architecture that was adapted for text classification, as shown in Figure 2.2. Through their work, Ebrahimi et al. (2016) realized that learning word representation directly within the model, rather than relying on generic pre-trained embeddings, like Word2Vec, allowed the CNN model to capture contextual relationship between words and phrases that may indicate grooming behavior more efficiently [ESO16]. The authors concluded that deep CNNs provide a promising direction for improving automated detection of online predators and suggested that future research could explore other deep learning architectures, such as Recurrent Neural Networks for further improvements in this domain.

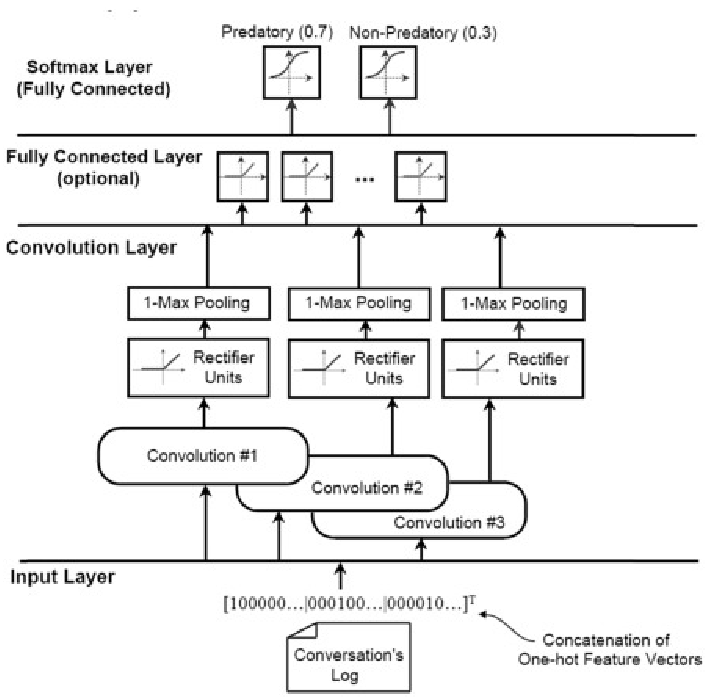


Figure 2.2: The proposed CNN Architecture used for OPI (Source:[ESO16])

Taking into consideration that grooming is a malicious act that happens over a period of time and with gaps in between its stages, the classification of an entire conversation is not reliable for a real life scenario, so the necessity of an early detection method was clear. Vogt et al. (2021) introduced the early Sexual Predator Detection (eSPD) method, which employs a two-tier approach using BERT models [MV21]. The system uses a local tier to classify message windows and a global tier to track a "skepticism" threshold, raising an alert only when suspicious activity is consistently detected [MV21]. By testing various BERT variants, such as Large and Base, the researchers demonstrated that these transformer-based models can balance the need for early warnings with high accuracy, representing a major step toward real-time preventative surveillance [MV21]. All of this was done on a new dataset created by Vogt et al. (2021) call PANC, which is a combination between the PAN12 and ChatCoder2 datasets (Figure 2.3).

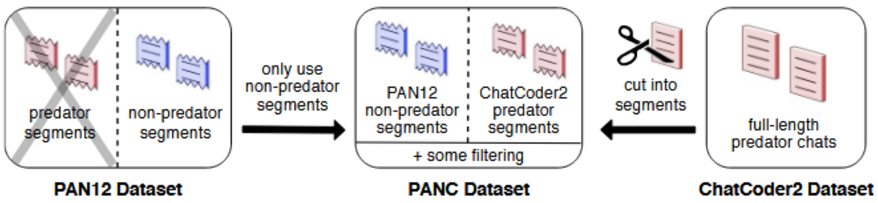


Figure 2.3: PANC dataset from PAN12 and ChatCoder2 dataset (CC2) (Source:[MV21])

2.4 Behavioral and Emotion-Based Approaches

The next logical approach to this problem was understanding the psychology behind online grooming. Gupta et al (2012) aimed to advance state-of-the-art in non-topical text classification models by discussing the first stem towards this system, characterize/profile online grooming theory from Psychology, specifically identify the indicators that characterize the various stages of online grooming [GKS12]. With the help of a professional psychologist, they manually annotated chat logs into six distinct stages: friendship forming, relationship forming, risk assessment, exclusivity, sexual and conclusion. Using Linguistic Inquiry and Word Count (LIWC) and logistic regression, they concluded that in a pedophile conversation relationship forming is the most prominent stage out of the six stages of online grooming, covering 40% of lines in a pedophile conversation, while sexual stage only covers around 24% of the lines [GKS12]. Furthermore, pedophiles do not wait for the end of the grooming process to approach a child to meet in person, on the contrary, they show that conclusion stage should be considered as a basic and recurrent stage like the other five stages [GKS12]. Ultimately, this work proved that predatory intent often occurs through non-sexual subtext, focusing on forming a relationship with the victim and recurrently bringing up the conclusion stage/the proposal to meet and showing the necessity to monitor deep linguistic patterns.

Wani et al. (2021) expanded on the analysis of behavioral traits by treating communication styles as Social Behavior Biometric (SSB) features [WAB21]. Their methodology utilized a hybrid feature extraction process that combined a specialized vocabulary model, PreVicVocab, with the MoodBook lexicon to quantify 11 emotional attributes [WAB21]. This allowed the system to differentiate between the more formal language used by predators and the informal, slang-heavy styles typical of younger victims [WAB21]. The study's analysis of eight distinct emotion categories revealed that predators often display significantly higher ranges of trust and anticipation as they work to manipulate a victim's confidence [WAB21]. By integrating these social biometric traits into a Random Forest classifier, the researchers achieved a peak F_1-score of 0.95, as shown in Table 2.4, demonstrating that combining emotional cues with vocabulary-based features provides a highly accurate method for identifying grooming nuances [WAB21].

Dataset	Accuracy(DT)	F_1 score(DT)	Accuracy(SVM)	F_1 score(SVM)	Accuracy(RF)	F_1 score(RF)
D1	99.76	0.89	95.34	0.57	99.35	0.81
D2	99.86	0.94	99.69	0.86	99.88	0.95

Table 2.4: Performance of Several Supervised Learning Techniques using Vocabulary and Emotion-based Features (Source:[WAB21])

Tabuyo (2024), much like Wani et al. (2021), decided to analyze the emotional subtext in predatory conversations. Combining pretrained emotion classification models (such as RoBERTa and ModernBERT fine-tuned on the GoEmotions dataset) with binary-labeled conversations from the PAN12 corpus, he transformed the chat logs into multi-label emotion vectors [Tab24]. A critical advance in this research is the application of SHAP (SHapley Additive exPlanations), an explainable AI technique used to interpret the decisions of the XGBoost classifier, to identify the top 10 most influential emotions for grooming prediction [Tab24]. As a complement to SHAP analysis, a method was used that calculates the mean activation of each emotion. Filtering out low-intensity activations and neutral emotions, Tabuyo (2024) was able to focus on the top 5 emotions with the highest mean [Tab24]. In the end, the research showed that curiosity, diversion, love, sadness and optimism were the predominate emotions in predatory conversations. Table 2.5 reports the performance of the models restricted to these emotions.

Model	F1-score micro	F1-score macro	F1-score weighted
SamLowe/roberta-base-go-emotions	0.5530	0.5257	0.5525
fyaronskiy/ModernBERT-large-englsh-go-emotions	0.5886	0.5614	0.5866
sangkm/go-emotions-fine-tuned-distilroberta	0.5530	0.5257	0.5525

Table 2.5: Results of the pretrained models (Source:[Tab24])

2.5 Conclusions and comparisons between the presented methods

Ref.	Authors	Year	Method Type	Model	Feature Type	Detection Type	Dataset	Results
[Tab24]	Tabuyo	2024	Deep Learning	RoBERTa / ModernBERT emotion classifiers	Emotion vectors (GoEmotions), contextual embeddings	Conversation classification	PAN12	Best F_1 (micro): 0.5886
[MV21]	Vogt et al.	2021	Deep Learning	BERT (large, base, MobileBERT)	Contextual embeddings	Early detection using sliding windows	PANC (PAN12 + ChatCoder2)	Best model BERT Base F_1: 0.89
[EVTJ+13]	Escalante et al.	2013	Machine Learning	Classifier Chain	Bag-of-Words	Stage-based conversation classification	PAN12	Overall F_1: 78.98%
[VTJGE+12]	Villatoro-Tello et al.	2012	Machine Learning	Neural Networks / SVM	Bag-of-Words, TF-IDF	Two-stage detection (conversation + user)	PAN12	F_1 = 0.8734, $F_{0.5}$ = 0.9346
[ESO16]	Ebrahimi et al.	2016	Deep Learning	Convolutional Neural Network (CNN)	Word embeddings, BoW, one-hot	Conversation classification	PAN12	Best F_1 = 80.87%
[WAB21]	Wani et al.	2021	Machine Learning	Random Forest / SVM / Decision Tree	Vocabulary + emotion lexicon (MoodBook)	Behavioral conversation analysis	PAN12	Accuracy 99.88%, F_1 = 0.95

Table 2.6: Summary of Related Work.

By analyzing Table 2.6 we end up with very insightful information.

- Early approaches relied mainly on classical ML, like Support Vector Machine, Neural Networks, Classifier Chains, while later work introduced deep learning architectures such as CNNs and BERT.
- The methods modeling conversation structure or stages capture grooming progression better than simple global classifiers.
- Recent studies show that emotional patterns and behavioral signals can improve predator detection by capturing psychological cues in conversations.
- Deep learning models generally achieve strong performance, especially when contextual information is used. However, hybrid approaches combining linguistic + emotional features with classical ML can also achieve very high accuracy.

Chapter 3

Theoretical Foundations

This chapter presents the theoretical background that stands behind the methods that are used in this work. It is organized into two main areas. The first covers the core principals of Deep Learning and Supervised Learning, training mechanics and regularisation strategies. The second addresses the foundation of Natural Language Processing. Together, these two provide the conceptual framework on which the proposed approach is built.

3.1 Deep Learning and Supervised Learning

Supervised learning is a machine learning approach in which a model learns from labeled data, meaning each training example is associated with a known correct output. During training, the model compares its predictions with the true labels and adjusts its parameters to minimize prediction errors, ultimately learning a mapping from inputs to outputs that generalizes to unseen data. [Nas17]

Deep Learning is a specialized subset of Machine Learning that focuses on training neural networks, with multiple hidden layers, to learn hierarchical feature representation directly from observational data. [Nie25] In many practical applications, deep learning models are trained using the supervised learning paradigm, where the label examples guide the learning process and enable the network to improve its predictive performance through iterative optimization.

Binary classification is a fundamental supervised learning task where the goal is to predict a target variable y drawn from two discrete classes, typically denoted as $\{0,1\}$. The standard architecture used for this kind of task is the Multilayer Perceptron(MLP), a type of neural network where information moves in one direction (from the input layer through one or more hidden layers and finally to the output). Each layer builds on the one before it, gradually transforming the raw input into representations that are rich enough for the network to make reliable prediction.[Agg20]

What makes this layered structure powerful is the use of activation functions, which introduce the non-linearity that allows the network to go beyond simple linear transformations and model more complex relationships in the data. Two of the most widely used are Sigmoid and ReLU. The Sigmoid function

$$\sigma(v) = \frac{1}{1 + e^{-v}}$$

maps any input to a value between 0 and 1, which makes its output easy to interpret as a probability. However, it tends to struggle when inputs are very large or very small, producing near-zero gradients that slow down or even stall the learning process. ReLU

$$\Phi(v) = \max(0, v)$$

is simpler, faster to compute and does a much better job of keeping gradients alive during training. [Sza20]

Training, in essence, is fundamentally an optimization problem: finding a set of weights W that minimizes a loss function - the quantified mismatch between the network's prediction ŷ and the ground truth label y. For binary task, BCEWithLogitsLoss is the preferred choice, combining Sigmoid and Binary Cross-Entropy into a single numerically stable step, with the added mathematical advantage that $\sigma'(z)$ term cancels out during gradient computation. [Nie25] Once the loss is computed, backpropagation walks backwards through the network using the chain rule to calculate the gradient of the loss with respect to every parameter. These gradients are then used by an optimizer to update the weights in the direction that reduces the loss. [RDGC95]Modern optimizers like Adam and AdamW improve on basic gradient descent by building up momentum across updates and adapting the learning rate for each parameter individually, leading to faster and more stable convergence.

3.2 Natural Language Processing

Natural Language Processing is a computerized approach to analyzing and representing naturally occurring texts with the goal of achieving human-like language understanding. The field evolved from rule-based systems toward statistical and then deep learning models, with the consistent objective of generalization - identifying regularities in language rather than memorizing training data.

Before text can be processed by nerual networks, it must undergo tokenization (the segmentation of input strings into smaller units such as words or subwords). [Lid01]Methods like Byte Pair Encoding (BPE) and WordPiece manage the trade-off between sequence length and data sparsity by merging frequently co-occurring characters into subword tokens. [XZ23] Earlier approaches such as one-hot encoding and TF-IDF vectorization, while foundational, lacked the semantic depth required for complex language tasks, which led to the development of word embeddings such as Word2Vec and GloVec, dense, low-dimensional representations learned by predicting words from their surrounding context. However, these static embeddings assign the same representation to a word regardless of usage, failing to distinguish, for example, between "right turn" and "right choice". This limitation prompted the shift towards contextual embeddings, where representations are dynamically adjusted based on the surrounding sequence.[Agg20]

The Transformer replaced recurrent neural networks with a model based entirely on attention mechanisms, with the key advantage of high parallelization. Its core innovation, the Self-Attention Mechanism, learns direct intersections between every pair of inputs in a sequence through a QKV model (transforming each input into a Query(Q), Key(K) and Value(V) vector). Multi-head Self-Attention extends this by attending to different linguistic features simultaneously across multiple sub-spaces, greatly enriching the resulting representations. [XZ23] Building on this, BERT utilities the Transformer encoder and is pre-trained through Masked Language Modeling (MLM) and next-sentence prediction, capturing context from both sides of a token simultaneously. RoBERTa refines this through more robust training strategies, improving performance across a wide range of tasks. Both models exemplify the paradigm of transfer learning, where a model pre-trained on a vast amounts of text is subsequently fine-tuned on task-specific labeled data by adding a small output layer, ensuring that rich pre-trained data by adding a small output layer, ensuring that rich pre-trained representations are effectively leveraged for precise downstream classification.[XZ23]

Chapter 4

Individual Contribution

This chapter presents the main contributions developed within this thesis for the early detection of online grooming conversations. First, the preprocessing pipeline used to transform the PAN12 dataset into a machine learning-ready format is described. Subsequently, several classification approaches are introduced, ranging from a traditional TF-IDF and Logistic Regression baseline to transformer-based models such as DistilBERT and RoBERTa. The chapter also presents the integration of emotional information through the GoEmotions model and the proposed feature fusion strategies that combine semantic and emotional representations. Finally, the implementation details of the resulting fusion models are discussed, highlighting the design choices adopted to improve early grooming detection performance.

4.1 Data Processing

The PAN12 Sexual Predator Identification dataset was preprocessed to transform the original XML chat logs into a format suitable for machine learning. Each conversation was parsed to extract the author and text of every message, while messages with missing or empty content were removed. Conversations were then labeled ("1" if at least one participant was identified as predator and "0" otherwise) using the predator lists provided with the dataset. To preserve conversational structure while keeping users anonym, original participant identifiers were replaced with conversation local speaker tags (SPK1, SPK2 etc.). Messages were labeled by speaker and kept in their original conversation order.

To support early detection experiments, multiple conversation snapshots were generated form each conversation at different stages of its progression. This enables the models to learn behavioral patterns from incomplete conversations and assess the presence of predatory activity as intersections unfold. Each generated sample contains the conversation text, its label, the conversation identifier and the relative progress within the conversation. The resulting samples were stored in CSV format and used for training and evaluating the proposed models. Figure 4.1

```
label,progress,conv_id,text,window_strategy
0,0.0967741935483871,0,"SPK1: Hola.
SPK2: hi.
SPK2: whats up?
SPK1: not a ton.
SPK1: you?
SPK2: same.  being lazy.  M or f?",full
```

Figure 4.1: Processed data in CSV format

4.2 Baseline: TF-IDF and Logistic Regression

As a baseline approach for online grooming detection, a traditional Natural Language Processing (NLP) pipeline based on Term Frequency-Inverse Document Frequency (TF-IDF) features and Logistic Regression was implemented. This model serves as a reference point against which the performance of more advanced deep learning architectures can be compared.

The first stage converts each conversation window into a numerical representation using TF-IDF vectorization. This assigns a weight to each term according to its frequency within a document while reducing the influence of terms that frequently occur across the entire corpus [RW08], basically it allows the model to focus on words and phrases that are more informative for distinguishing between predatory and non-predatory conversations. The vectorizer was configured to use a maximum vocabulary size of 20,000 features and to extract both unigrams and bigrams. Terms appearing in fewer than two documents were removed, while terms occurring in more than 90% of the documents were ignored. Additionally, common English stop words were excluded to reduce noise in the feature space.

In the second stage the conversation windows transformed into TF-IDF vectors were used to train a Logistic Regression classifier. This estimates the probability of a sample belonging to the positive class through the sigmoid function [PLI02]. Given an input feature vector (x), the model computes $P(y = 1 \mid x) = \sigma(w^T x + b)$, where (w) represents the learned feature weights, (b) is the bias term and σ denotes the sigmoid activation function. The balanced class weights were used, due to the severe class imbalance present in PAN12 so a grater importance would be assign to the minority class during optimization.

The baseline model is computationally efficient, easy to train and highly interpretable, as it allows the identification of words and phrases that contribute most to the classification decision. However, it relies solely on word occurrence statistics and therefore cannot effectively capture semantic meaning, conversational context, or long-range dependencies that may be present in grooming conversations.

4.3 DistilBERT Model

To improve upon the limitations of traditional lexical features, a transformer-based approach using DistilBERT was implemented. DistiBERT is a compressed version of BERT that retains much of the original model's language understanding capabilities while requiring fewer parameters and computational resources [SDCW19]. This makes it particularly suitable for tasks involving large conversational datasets.

The model uses the pretrained distilbert-base-uncased encoder to transform each conversation window into a contextualized vector representation. Prior to encoding, the text is tokenized, padded and truncated to a maximum length of 128 tokens. The encoder operates in inference mode, meaning that its pretrained weights remain frozen throughout the training process and are not updated during optimization. For each input sequence, the representation of the first token, commonly referred to as the CLS (Classification token for sentence-level tasks) token, is extracted from the final hidden layer and serves as a summary of the entire input sequence and is represented by a 768-dimensional embedding vector. These embeddings capture contextual and semantic information learned during pretraining on large-scale text corpora. The generated embeddings are then used as input to a binary classification model. The classifier consists of a single fully connected layer that maps the 768-dimensional embedding to a single output neuron, followed by a sigmoid activation function. The output represents the probability that the conversation window belongs to the predatory class.

The model was trained using a weighted Binary Cross-Entropy loss function, with positive samples weighted according to the class imbalance in the training set and optimized with the Adam optimizer using a learning rate of $1e-3$. Training was performed for eight epochs with a batch size of 32. Since the DistilBERT encoder remains frozen, only the parameters of the classification layer are updated during training. This significantly reduces training time while still benefiting from the rich semantic representations produced by the pretrained language model. By leveraging contextual embeddings rather than sparse lexical features, DistilBERT is able to capture semantic relationships, conversational context and linguistic patterns that are difficult to model using traditional machine learning approaches.

4.4 RoBERTa Model

To further investigate the impact of transformer-based language models on grooming detection, a second architecture based on RoBERTa was evaluated. RoBERTa (Robustly Optimized BERT Approach) extends the original BERT training methodology through improved pretraining procedures and larger-scale training data, resulting in stronger language representations [LOG+19].

The implementation uses the pretrained roberta-base model as a feature extractor. Compared to DistilBERT, RoBERTa contains approximately 125 million parameters and processes input sequence of up to 256 tokens, allowing a larger portion of each conversation window to be preserved during encoding. This is particularly relevant for online grooming detection, where important behavioral cues may emerge gradually throughout the dialogue. The extracted RoBERTa embeddings are used as input to the same linear classification architecture described in the previous section. However, several modifications were introduced during training to better address the characteristic of the PAN12 dataset.

First, class imbalance was handled through a weighted Binary Cross-Entropy loss, where positive samples receive a larger penalty than negative samples. This encourages the model to place greater emphasis on correctly identifying predatory conversations, which represent only a small fraction of the dataset.

Second, the decision threshold was not fixed at 0.5. Instead, multiple threshold values were evaluated on the validation set and the threshold producing the highest F1-score was selected. This allows the classifier to achieve a more suitable balance between precision and recall.

Finally, separate thresholds were learned for each early-detection checkpoint. Since the amount of available information differs significantly between the beginning and the end of a conversation, adapting the decision threshold to each progress level can improve detection performance throughout the dialogue.

The model was trained for eight epochs using the Adam optimizer with a learning rate of $1e-3$. The checkpoint achieving the highest validation F1-score was selected as the final model and stored together with the optimized threshold values. Overall, RoBERTa introduces a longer context window, explicit for the handling of class imbalance and validation-based threshold optimization. These additions make the model particularly suited to the early detection scenario explored in this work.

4.5 GoEmotions

In addition to the semantic representations generated by DistilBERT and RoBERTa, an emotion-based feature extraction approach was implemented using the pretrained

SamLowe/roberta-base-go_emotions model. This model is fine-tuned on the GoEmotions dataset and is designed to identify the presence of multiple emotion categories within a text [DMAK+20].

Each conversation window is tokenized and processed by the pretrained model parameters remaining frozen throughout the process. The output of the network consists of a set of logits corresponding to the emotion categories learned during training. A sigmoid activation function is then applied to these logits to obtain independent probability scores for each emotion. The resulting representation is a 28-dimensional feature vector, where each dimension corresponds to the predicted probability of a specific emotion. Unlike the contextual embedding produced by DistilBERT and RoBERTa, these features provide an emotional profile of the conversation window rather than a semantic representation of its content. To reduce computation time, the extracted emotion features are cached and reused whenever the same conversation window is processed again.

As discussed above, the resulting 28-dimensional emotion vectors provide an emotional profile of each conversation window and emotion information alone is insufficient in the task of early detection of grooming behavior. While the model can identify emotional signals such as curiosity, fear or desire, it does not capture the semantic content of the conversation or the strategic interaction patterns typically used by predators.

Grooming conversations often rely on subtle manipulative behaviours that develop over time and cannot be accurately represented solely through emotional categories. As a result, the GoEmotions model is better suited as a complementary source of information rather than a standalone solution. Its emotional representations can enrich transformer-based semantic embeddings, but they do not provide enough information on their own to reliably distinguish predatory from non-predatory conversations.

4.6 Feature Fusion

The previous experiments showed that transformer-based models and emotion-based models capture different aspects of conversations. To leverage both sources of information simultaneously, a feature fusion approach was implemented.

The principle of feature fusion is based on concatenating the feature vectors generated by two independent models into a single representation. Let $x_{encoder}$ denotes the contextual embedding produced by a transformer model and $x_{emotion}$ denotes the emotion vector generated by GoEmotions. The fused representation is defined as

$$x_{\text{fused}} = [x_{\text{encoder}} \parallel x_{\text{emotion}}]$$

,where (||) denotes vector concatenation. For the DistilBERT-based fusion, a 768-dimensional contextual embedding is combined with the 28-dimensional GoEmotions vector, resulting in a fused representation of 796 features. The same procedure is applied to RoBERTa embeddings, which also produce 768-dimensional vectors, leading to an identical fused dimensionality of 796 features.

The fusion process is performed after feature extraction. Each conversation window is independently processed by both models and the resulting vectors are concatenated to form a single feature vector containing both semantic and emotional information. This approach allows the classifier to learn from complementary sources of information without modifying the internal architecture of the pretrained models. Since generating transformer embeddings is computationally expensive, the fused feature vector is precomputed and stored locally. This caching strategy eliminates the need to repeatedly run the feature extraction models during training and evaluation, substantially reducing execution time while preserving the same feature representations. By combining contextual and emotional information into an unified representation, the fusion approach aims to provide a more complete description of grooming conversation than wither feature type could offer individually.

4.6.1 BERT and GoEmotions Fusion

The classifier used for the fused representation is a single linear layer that maps the 796-dimensional input to one output value. Instead of training this classifier entirely from scratch, the model is initialized using the previously trained DistilBERT classifier. The weights corresponding to the first 768 dimensions are copied from the BERT-only model, while the weights corresponding to the 28 GoEmotions features are initialized with zero. As a result, the fused model starts from the same semantic decision boundary as the original BERT classifier, although block-wise L2 normalization applied before classification means it does not reproduce BERT-only predictions exactly.

During training, the BERT-related weight columns are frozen, meaning that they are not updated by gradient descent. Only the weights associated with the GoEmotions features, together with the bias term, are trained. This makes the emotion component act as a correction layer on top of the already learned semantic classifier. In other words, the model keeps the decision patterns learned from DistilBERT and only adjusts them using additional emotional information. Before classification, the two feature blocks are normalized separately using L2 normalization. This step is important because the BERT embedding and the GoEmotions vector have different sizes and value distributions. By normalizing them independently, the model prevents the larger BERT representation from dominating the smaller emotional feature vector only because of its dimensionality.

The fused model was trained for six epochs using Binary Cross-Entropy loss and the Adam optimizer with a learning rate of $5e-3$. Both the model checkpoint and the decision threshold were selected on the validation set by choosing the configuration that maximized the F1-score. This fusion strategy allows the model to preserve the strong semantic information learned by DistilBERT while incorporating emotional cues from GoEmotions. In this way, GoEmotions does not replace the transformer representation, but complements it by adding emotional information that may support early grooming detection.

4.6.2 RoBERTa and GoEmotions Fusion

The RoBERTa and GoEmotions fusion model follows the same general feature concatenation principle described in the previous section. The main difference is found in the training strategy. While the BERT fusion model freezes the semantic part of the classifier and only trains the GoEmotions-related weights, the RoBERTa fusion model updates all feature weights jointly. This allows the classifier to adjust both the semantic and emotional contributions at the same time.

The model is initialized from the previously trained RoBERTa classifier, so the semantic part already starts from learned weights. However, unlike the BERT fusion approach, these weights are not frozen during training. A lower learning rate of $1e-3$ is used in order to avoid large updates that could damage the previously learned RoBERTa-based decision patterns. The training process also uses a weighted Binary Cross-Entropy loss to address the strong class imbalance in the dataset. Another important difference is the use of progress-specific thresholds, as previously done for the simple RoBERTa model.

Overall, this fusion strategy is more flexible than the BERT and GoEmotions fusion model. Instead of using GoEmotions only as a small correction over a fixed semantic classifier, the RoBERTa fusion model jointly adjusts semantic and emotional features, making it better adapted to the early detection setting.

Chapter 5

Experiments

This chapter presents the experimental evaluation of the models introduced in the previous chapter. The objective of the evaluation is to determine how effectively each approach can identify grooming behavior at different stages of a conversation. Since grooming is a gradual process that develops over times, evaluating models only on complete conversations would not accurately reflect a real-world detection scenario. To address this limitation, each model was evaluated at multiple conversatio progress levels corresponding to 10%, 20%, 40%, 60%, 80% and 100% of the available conversation. Performance was measured using the F1-score, which provides a balanced evaluation of precision and recall and is particularly suitable for highly imbalanced datasets such as PAN12.

5.1 Baseline Results

The TF-IDF and Logistic Regression model served as the baseline for all subsequent experiments. Although the model achieved reasonable performance, its reliance on lexical features limited its ability to capture semantic relationships and conversational context.

As shown in Figure 5.1, the baseline achieved and F1-score of approximately 0.67 across all conversation stages. While these results demonstrate that word occurrence patterns contain useful information for grooming detection, the relatively stable performance across increasing conversation lengths suggests that the model struggles to take advantage of additional contextual information. This behavior highlights one of the main limitations of traditional machine learning approaches for conversational analysis.

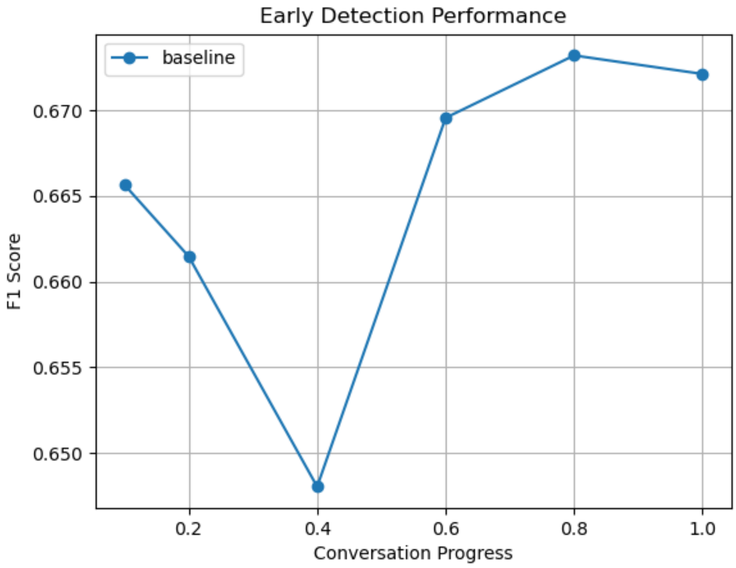


Figure 5.1: Baseline Results

5.2 DistilBERT and RoBERTa Results

5.2.1 DistilBERT

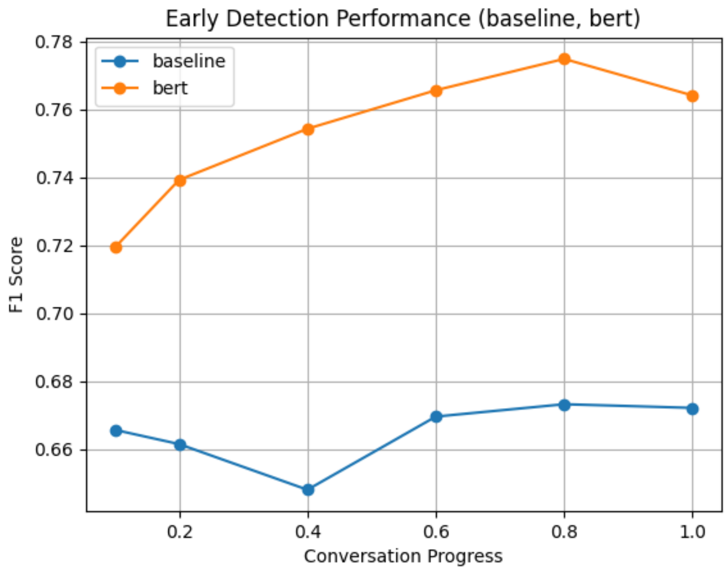
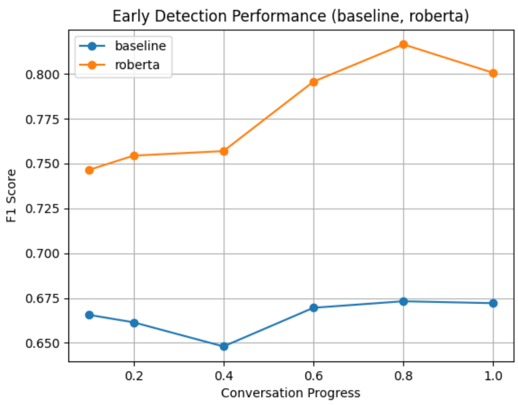


Figure 5.2: Baseline vs DistilBERT

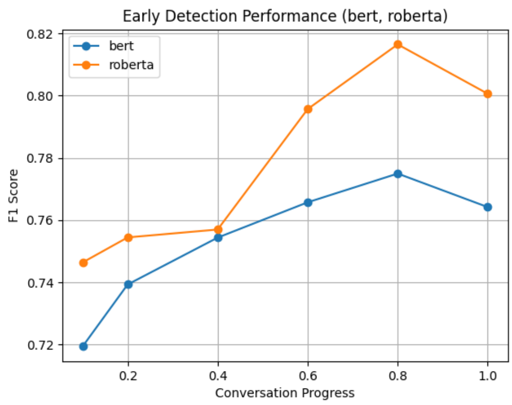
The introduction of contextual embeddings through DistilBERT produced a substantial improvement over the baseline model. Figure 5.2 The results show a consistent increase in performance across all conversation progress levels. The model achieved significantly higher F1-scores than the baseline, demonstrating the effectiveness of transformer-based language representations for grooming detection. Furthermore, performance generally improved as more conversation context became available, indicating that DistilBERT successfully captures behavioral patterns that emerge gradually throughout the dialogue.

5.2.2 RoBERTa

RoBERTa was evaluated under the same experimental conditions. Similarly as before, the transformer-based model, RoBERTa did significantly better than the baseline model proposed Figure 5.3a. Regarding the comparison with DistilBERT, the model outperforms the previous transformer, especially in the later stages Figure 5.3b. Ultimately, RoBERTa maintained strong performance throughout all stages of the conversation, demonstrating a consistent ability to leverage contextual information for grooming detection.



(a) Baseline vs RoBERTa



(b) RoBERTa vs BERT Results

5.3 Fusion Results

As mentioned previously in 4.5 the GoEmotion model is not suited as a standalone solution to this task, recording a F1-score of 0.028. That is way the fusion is proposed in this situation. The idea was to see if the transformer-based models would benefit from the emotional signal provided from this model.

5.3.1 DistilBERT and GoEmotions Fusion

For the first fused model we have the combination of DistilBERT and GoEmotions, that produced approximately 0.02% improvement over the standalone DistilBERT, as it can be seen in Figure 5.4. The gains were only during early stages of the conversation and model is weaker than the standalone version in later stages so the claim that emotional information can complement semantic representations cannot be fully supported with this fusion.

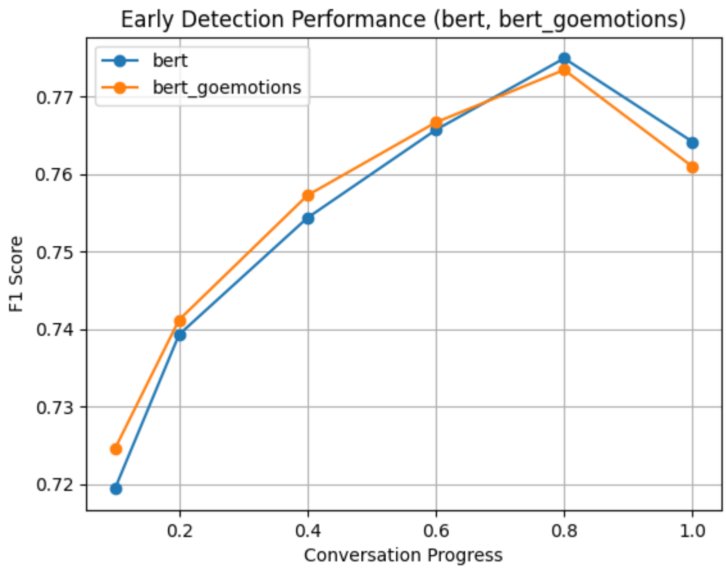


Figure 5.4: DistilBERT vs Fusion model based on DistilBERT

5.3.2 RoBERTa and GoEmotions Fusion

The fusion with RoBERTa was more successful. The GoEmotions model improved RoBERTa by 2% at 40% of the conversation and it keeps itself above the standalone RoBERTa results over the last stages of the conversation, as it can be seen in Figure 5.5, when the improvement between DistilBERT and its fusion was only of 0.02%. Despite the fact that for the first 20% of the conversation, the score was worse, from 20% to 60%, the emotional analysis from GoEmotions drastically raises the results. This proves that the combination of contextual language understanding and emotional profiling is a suitable approach in the task of early detection of online grooming.

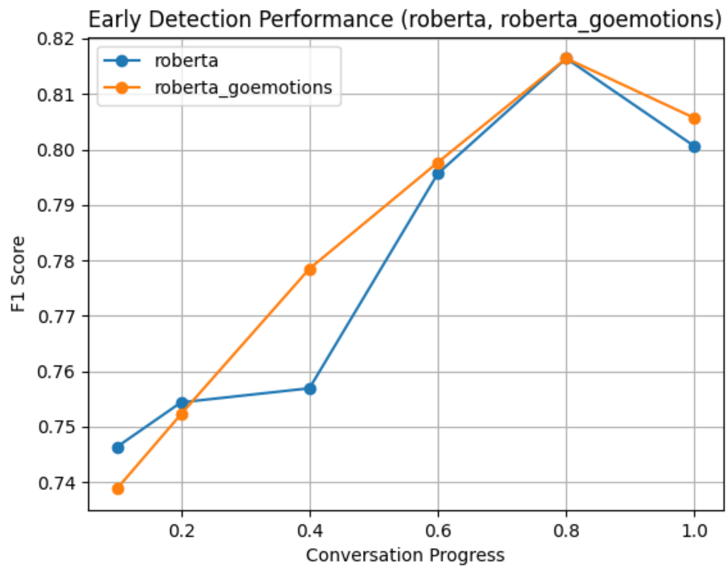


Figure 5.5: RoBERTa vs Fusion model based on RoBERTa

5.4 Error Analysis

To better understand the strengths and limitations of the evaluated models, an error analysis was performed on the PAN12 test set under two complementary settings. The first uses complete conversations, where each conversation is represented by a single instance at 100% progress (155,101 conversations, 3,723 positive). The second follows the early-detection evaluation protocol used throughout this thesis, where all cumulative conversation windows are evaluated (830,234 windows, 22,405 positive). For each setting, misclassified samples were extracted and grouped into false positives (safe conversations classified as predatory) and false negatives (predator conversations classified as safe). In addition to quantitative statistics, representative examples were manually inspected and assigned heuristic categories such as short conversations, casual openers, emotional interactions and explicit content to identify recurring error patterns.

A consistent pattern emerged across all evaluated approaches. The majority of errors were associated with short conversations containing limited contextual information. False positives frequently originated from benign conversations involving greetings, emotional language, affection, or casual social interactions that resemble grooming behavior at a surface level. Conversely, false negatives were commonly observed in conversations where grooming cues remained implicit, indirect, or insufficiently developed. These findings suggest that the availability of conversational context is a key factor influencing model performance and that early-stage grooming detection remains challenging regardless of the underlying architecture.

5.4.1 Baseline

The TF-IDF and Logistic Regression baseline exhibits a recall-oriented behavior in both evaluation settings. At conversation level, the model achieved an F1-score of 0.54 for the predator class, generating 4,614 false positives and 642 false negatives. In the window-level setting, performance improved to an F1-score of 0.65 with a recall of 90.6%, although a large number of false positives (19,348) remained.

Overall, the baseline demonstrates two key limitations: lexical shortcut learning and context blindness. The model relies heavily on keyword overlap and struggles to capture the broader conversational context, leading to frequent misclassifications of both benign and early-stage grooming conversations.

5.4.2 DistilBERT

For DistilBERT at the conversation level, it reached an F1-score of 0.665 for the predator class, with 1,431 false positives and 1,154 false negatives. At the window level, performance was higher, with an F1-score of 0.764, precision of 76.2% and recall of 76.7%.

The results suggest that DistilBERT is capable of identifying many contextual grooming signals, but its performance is still affected by conversations containing little textual information or highly implicit predatory behavior. The predominance of short conversations among both false positives and false negatives highlights the continued importance of conversational context for reliable grooming detection.

5.4.3 RoBERTa

The standalone RoBERTa model achieved an F1-score of 0.687 at conversation level, generating 1,174 false positives and 1,163 false negatives. Under the early-detection evaluation setting, the model reached an F1-score of 0.794, with 77.2% precision and 81.8% recall, indicating a stronger ability to identify grooming signals throughout conversation progression.

Overall, RoBERTa indicates the same errors in short conversations as DistilBERT does, but its performance from the previous is noticeable in both conversation and windows level analysis.

5.4.4 DistilBERT Fusion

The BERT + GoEmotions fusion model achieved an F1-score of 0.761 in the window-level evaluation, with a precision of 73.7% and a recall of 78.6%. The model generated 6,268 false positives and 4,796 false negatives, indicating a slightly stronger tendency to identify predator windows at the expense of additional false alarms.

Unlike the standalone DistilBERT model, the fusion architecture produces more recall-oriented predictions, detecting additional predator windows while also increasing the number of false positives. Overall, the results suggest that emotional information alters the model's decision behavior but does not lead to a meaningful improvement in overall detection performance. As with the previous models, the lack of sufficient conversational context remains the primary factor contributing to classification errors.

5.4.5 RoBERTa Fusion

The RoBERTa + GoEmotions fusion model achieved the strongest overall performance among the evaluated approaches. At conversation level, it obtained an F1-score of 0.689, producing 917 false positives and 1,286 false negatives. In the window-level setting, the model achieved an F1-score of 0.804, with a precision of 80.7% and a recall of 80.1%.

Overall, the results suggest that the integration of emotional features provides additional information that helps reduce the number of false alarms while maintaining strong detection performance. However, as with the previous models, conversations containing very little textual evidence remain the primary source of classification errors, highlighting the continued challenge of early grooming detection in short conversational exchanges.

5.5 Comparative Analysis

To facilitate a direct comparison between all evaluated approaches, Table 5.1 summarizes the obtained F1-scores across all conversation progress levels.

Progress	Baseline	GoEmotions	DistilBERT	RoBERTa	DistilBERT + GoEmotions	RoBERTa + GoEmotions
10%	0.666	0.010	0.719	0.746	0.724	0.738
20%	0.661	0.009	0.739	0.754	0.741	0.752
40%	0.648	0.014	0.754	0.756	0.757	0.778
60%	0.670	0.017	0.765	0.795	0.766	0.797
80%	0.673	0.019	0.774	0.816	0.773	0.816
100%	0.672	0.028	0.764	0.8005	0.760	0.805

Table 5.1: F1-score comparison of the evaluated models at different conversation progress levels

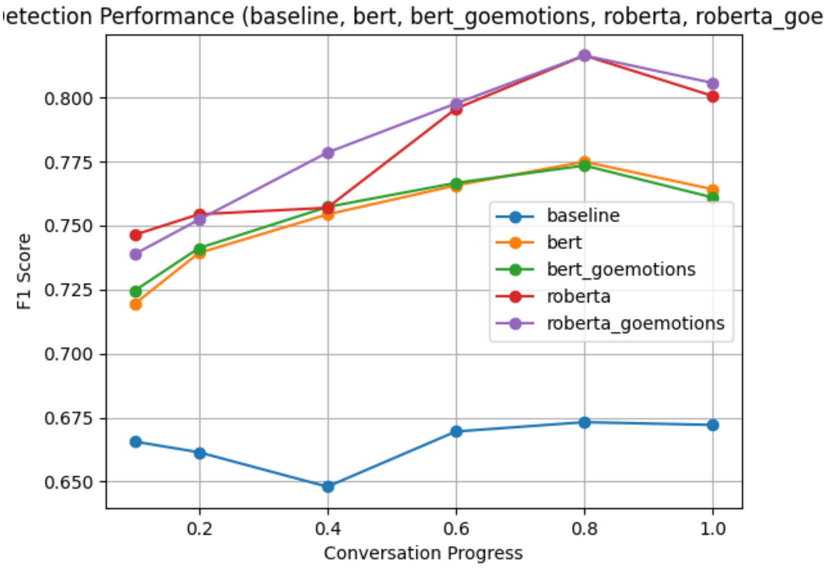


Figure 5.6: Comparison Between all the models

Several important observations can be made from these results. First, transformer-based approaches consistently outperform the traditional TF-IDF baseline, confirming the importance of contextual language representations. Second, the poor performance of the standalone GoEmotions model demonstrates that emotional information alone is not sufficient for grooming detection. Finally, the fusion models show that emotional signal can provide additional value when combined with semantic representations.

Among all evaluated approaches, the RoBERTa and GoEmotions fusion model achieved the best performance during the 20% to 60% of a conversation while maintaining competitive results on complete conversations. This is particularly important because the primary objective of grooming detection systems is to identify potentially harmful intersections as early as possible.

5.6 Comparison with Existing Approaches

To better asses the effectiveness of the proposed model, its performance was compared with several representative approaches from the literature. Although direct numerical comparisons should be interpreted with caution due to differences in datasets, evaluation protocols and experimental settings, such comparisons provide valuable insights into the strengths and limitations of the proposed approach.

Method	Dataset	F1-score	Recall	Early Detection
Villatoro-Tello et al. (2012)	PAN12	0.8734	0.7874	No
Escalante et al. (2013)	PAN12	0.6542	0.4991	Partial
Ebrahimi et al. (2016)	PAN12	0.8087	–	No
Wani et al. (2021)	PAN12	0.9500	–	No
Tabuyo (2024)	PAN12	0.5886	–	No
Proposed RoBERTa + GoEmotions	PAN12	0.805	0.801	Yes

Table 5.2: Results from representative approaches from the literature and my method (Numerical comparison is not well suited for this test case)

Among the approaches evaluated on the PAN12 dataset, Villatoro-Tello et al. [VTJGE+12] achieved an F1-score of 0.8734 and a recall of 0.7874 using carefully engineered linguistic features. These results demonstrate the effectiveness of domain-specific feature engineering for predator identification. However, despite obtaining a lower overall F1-score of 0.805, the proposed model achieved a higher recall of 0.801. From a practical perspective, this is an important advantage, as recall measures the ability to correctly identify predatory conversations. In child protection applications, failing to detect a predator is considered more costly than generating a false positive alert.

Escalante et al. [EVTJ+13] reported an F1-score of 0.6542 and a recall of 0.4991 using classifier chains designed to model grooming stages. While their work demonstrated the importance of considering grooming as a progressive process, the proposed model substantially improves both metrics. In particular, the recall increases from approximately 50% to over 80%, indicating a considerably stronger ability to identify predatory interactions.

Ebrahimi et al. [ESO16] achieved an F1-score of 0.8087 using convolutional neural networks. This result is slightly higher than the 0.805 obtained by the proposed model. Nevertheless, the difference is relatively small, while the proposed approach additionally incorporates emotional information and was explicitly evaluated in an early-detection setting. This suggests that the proposed model remains competitive despite addressing a more challenging objective.

Wani et al. [WAB21] reported the highest overall performance among the reviewed studies, with an F1-score of 0.9500 and an accuracy of 99,88%. These results indicate that their feature engineering strategy was highly effective for the specific dataset and evaluation protocol used. However, the study focused on full-conversation classification and did not evaluate performance at intermediate conversation stages. Consequently, it is difficult to asses how well the approach would perform in a real-time monitoring approach where only a fraction of the conversation may be available.

Vogt et al. [MV21] achieved an F1-score of 0.8900 using the eSPD framework. Their work is particularly relevant because it also addresses early detection. However the experiments were conducted on the PANC dataset, which was created to deal with the problem of class imbalance that the PAN12 dataset presents. Despite the lower overall F1-score obtained in this thesis, the proposed model follows a similar objective while additionally incorporating emotional information.

Tabuyo [Tab24] reported an F1-score of 0.5886 using emotion-based representations derived from GoEmotions. Compared to the proposed model's F1-score of 0.805, the difference is substantial. These results suggest that emotional information alone is insufficient for reliable grooming detection. Nevertheless, the improvement observed when emotional features are fused with contextual transformer embeddings confirms that emotions provide useful complementary information when combined with semantic representations.

Overall, the proposed RoBERTa and GoEmotions fusion model does not achieve the highest overall F1-score among all reported approaches. However, it achieves competitive performance while maintaining a high recall of 0.801 and supporting early detection. The results indicate that the fused model provides a balanced solution capable of identifying grooming behavior at different stages of a conversation, making the idea suitable for practical deployment in online safety systems.

Chapter 6

System Implementation and Deployment

This chapter presents the final system developed around the proposed grooming detection model. While the previous chapters focused on model design and experimental evaluation, here is also described how the selected model was integrated into a usable web-based platform. The system consists of a browser client, a FastAPI backend, a deployed RoBERTa and GoEmotions prediction pipeline and a secure communication layer for protecting sensitive conversation data.

6.1 Web-Based Client Application

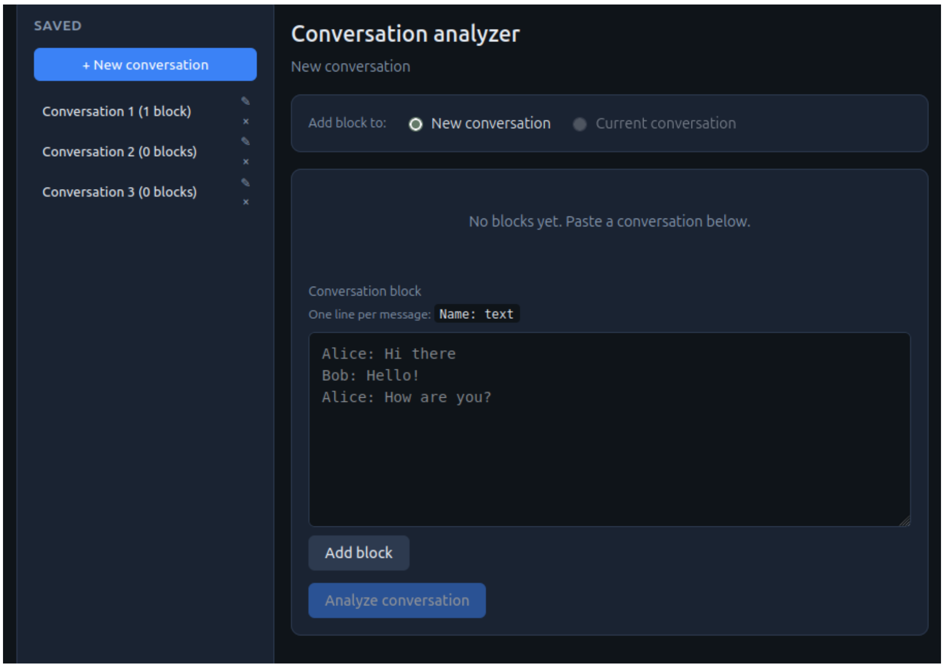


Figure 6.1: Application's web interface

The client application was implemented as a lightweight single page web interface using HTML, CSS and vanilla JavaScript. No fronted framework such as React or Vue was used, which keeps the application simple, portable and easy to serve directly from the backend. The static files are exposed by the FastAPI server at the `/app/` route, meaning that the user accesses the application through the browser rather than by opening the HTML file locally.

The role of the client application is to allow users to enter chat conversations, store them locally and submit them for analysis only when requested. Conversations are introduced in a simple structured format, where each line follows the pattern `Name: message`. This format is close to the representation used during preprocessing and allows the system to preserve both the author and the text of each message.

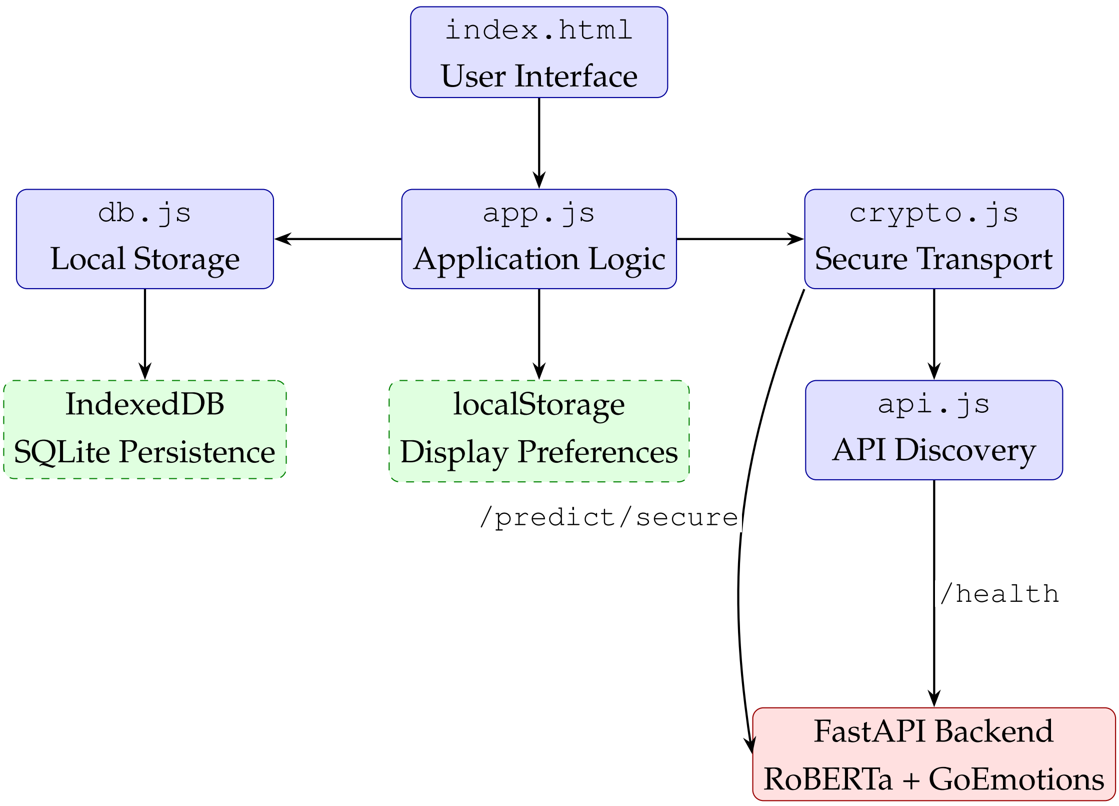


Figure 6.2: Architecture of the web-based grooming detection platform.

The user then adds one or more conversation blocks. Each block is parsed and stored locally. Once the conversation is ready, the user explicitly requests analysis. Only at this point are the messages transmitted to the server. After the backend processes the conversation, the result is displayed in the interface as either clear or flagged, together with the model score and the most contributing messages.

6.2 Local Conversation Storage

A privacy-oriented design choice of the application is that conversations are stored locally in the browser before being analyzed. This is implemented using SQLite through the `sql.js` library, with persistence provided by IndexedDB. As a result, users can create and manage conversations without requiring a remote database.

The local data model is composed of conversations and blocks. A conversation represents a complete chat session, while a block represents an ordered group of messages added by the user. This block-based structure supports incremental conversation construction, which reflects the early-detection scenario where new messages may become available over time.

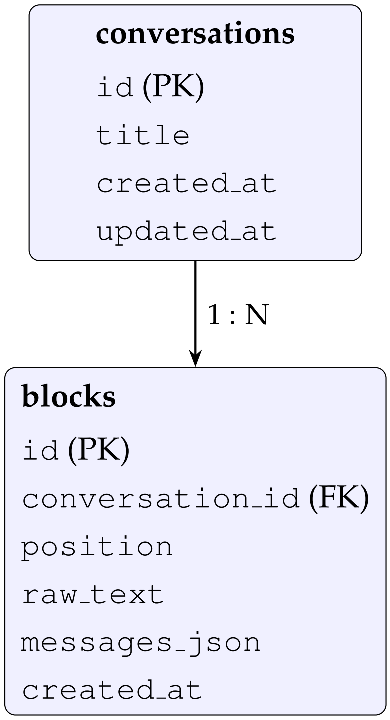


Figure 6.3: Client-side SQLite database schema.

Each block stores both the raw text entered by the user and a parsed JSON representation of the messages. The parsed format contains the author and text of each message, allowing the frontend to display the conversation clearly and the backend to receive structured input for prediction.

6.3 Deployment of the Proposed Model

The selected RoBERTa and GoEmotions fusion model was deployed through a FastAPI inference server. The backend loads the trained model during application startup and exposes prediction endpoints that can be accessed by the web client. This transforms the model from an experimental component into a usable service capable of analyzing conversations in real time.

When the user clicks the analysis button, the client gathers all messages from the current conversation and sends them to the backend. On the server side, the messages are converted into the same speaker-aware textual representation used during training. This ensures consistency between the experimental pipeline and the deployed system.

The backend returns several pieces of information after inference: the binary label, the predatory probability score, the number of analyzed messages and the name of the model used. When the model classifies a conversation as predatory, the system also returns the messages that contributed most strongly to the prediction. This message attribution mechanism provides an additional interpretability layer and can help moderators understand which parts of the conversation influenced the decision.

The deployed system uses the RoBERTa and GoEmotions fusion model as its prediction engine. This choice was motivated by the experimental results, where this model achieved the strongest early-detection performance while maintaining competitive full-conversation performance. Therefore, the final system integrates both semantic information from RoBERTa and emotional information from GoEmotions.

6.4 User Workflow

The main workflow of the application starts with initialization. During this step, the client detects the API endpoint, initializes local storage and establishes the secure communication session. After initialization, the user can create a new conversation or continue editing an existing one.

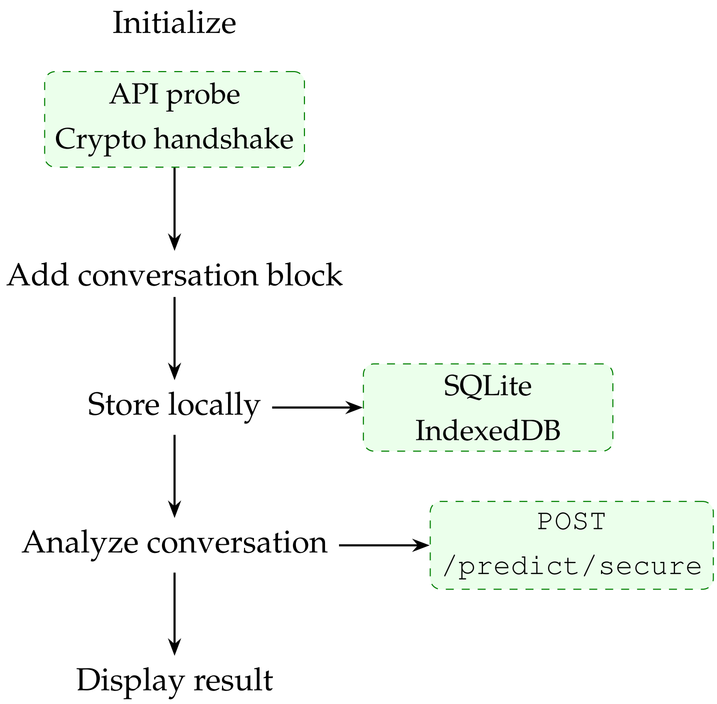
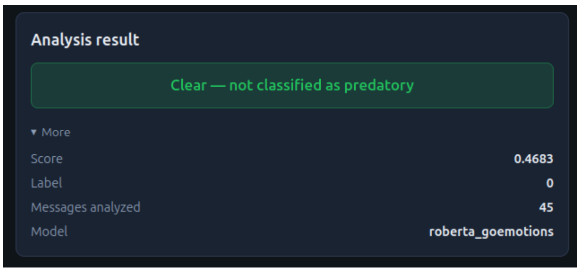
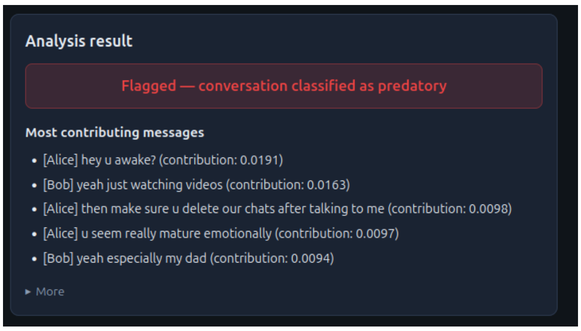


Figure 6.4: Main user workflow in the client application.

The user then adds one or more conversation blocks. Each block is parsed and stored locally. Once the conversation is ready, the user explicitly requests analysis. Only at this point are the messages transmitted to the server. After the backend processes the conversation, the result is displayed.



(a) Clear Response



(b) Flagged Response

6.5 Secure Communication and Privacy Protection

Since the system processes potentially sensitive conversations, privacy and secure communication were treated as important design requirements. The application uses a defense-in-depth approach based on two layers of protection.

The first layer is transport-level encryption through HTTPS. For local development, the project includes support for generating self-signed TLS certificates, allowing the server to run over HTTPS when TLS is enabled. This protects communication between the browser and the backend at the network level.

The second layer is application-level encryption. Before prediction requests are sent, the browser and server perform an Elliptic Curve Diffie-Hellman key exchange using the P-256 curve. The shared secret is then processed through HKDF-SHA256 to derive a session key. This key is used to encrypt prediction requests and responses with AES-256-GCM.

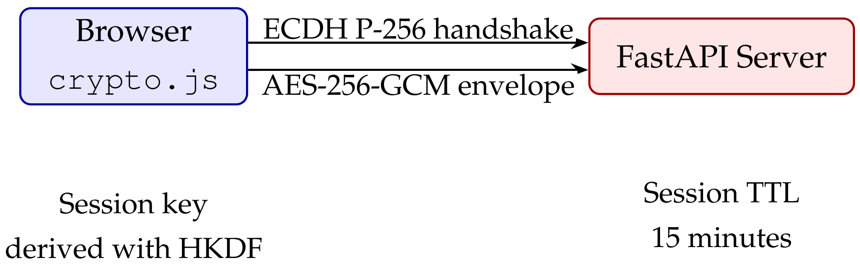


Figure 6.6: Security model based on TLS and application-level encryption.

This means that conversation messages are not sent as plaintext JSON over the network. Instead, the client sends an encrypted envelope to the `/predict/secure` endpoint. The server decrypts the request, performs prediction and encrypts the response before returning it to the client.

Each secure session is temporary and has a limited lifetime. This reduces the risks associated with long-lived cryptographic material. In addition, the plaintext prediction endpoint is disabled by default, encouraging the use of the secure prediction route during normal operation.

Chapter 7

Conclusions

This thesis investigated the problem of early online grooming detection through the combination sematic language representations and emotional information extracted from conversational data. Given the increasing prevalence of online communication platforms, the serious risk posed by online predators, the development of automated systems capable of identifying grooming behavior before conversations fully developed represents an important research challenge.

Several approaches were evaluated , ranging from a traditional TF-IDF and Logisstic Regression baseline to transformer-based models and hybrid fusion architectures. The experimental results confirmed that transformer-based language models significantly outperform traditional machine learning techniques by leveraging contextual representations capable of capturing subtle linguistic patterns associated with grooming behavior. Furthermore, the experiments demonstrated that emotional information alone is insufficient for reliable grooming detection, achieving poor performance when used as a standalone representation.

The main contribution of this work was the development and evaluation of feature fusion architecture that combine transformer-derived semantic embeddings with emotional representations generated by the GoEmotions framework. The results showed that emotional information provides complementary signals when integrated with contextual language representations. Among all evaluated approaches, the RoBERTa And GoEmotions fusion model achieved the strongest overall performance, particularly during the earliest stages of a conversation, demonstrating the effectiveness of combining semantic and emotional analysis for grooming detection.

In addition to the experimental evaluation, a complete web-based application was designed and implemented around the proposed model. The system allows users to analyze conversations through an intuitive interface and demonstrates how the proposed approach can be integrated into practical online safety tools.

Overall, the obtained results indicate that hybrid models combining language understanding and emotion analysis constitute a promising direction for early online grooming detection. The proposed approach achieved competitive performance while maintaining a strong focus on early identification, an essential requirement for systems intended to support child protection in online environments.

7.1 Future Work

Although the proposed approach achieved promising results, several directions remain open for future research and improvement.

A first limitation concerns the dataset itself. The PAN12 corpus remains one of the most widely used benchmarks for grooming detection, however, it was released in 2012 and does not fully reflect the language, platforms and interaction patterns observed in modern online communication and also poses class imbalance with the positive sample being a small percentage of the total train and test corpus. Future work should evaluate the proposed architecture on additional and more recent datasets in order to assess its ability to generalize beyond PAN12. Furthermore, extending the system to multilingual environments, would provide valuable insights into transferability of semantic and emotional representations across languages.

The early detection methodology can also be further refined. While this thesis evaluates models at fixed conversation progress checkpoints, future research could investigate dynamic decision policies that adapt detection thresholds according to conversation progress. Such approaches may improve the trade-off between early intervention and false positive alerts. Additional studies could also explore alternative windowing strategies, such as sliding windows or recency-weighted message representations, to better capture grooming behaviors that emerge during specific stages of a conversation.

From a modeling perspective, the proposed fusion architecture relies on the combination of contextual embeddings and emotional features. Future work could incorporate additional sources of information, including dialogue structure characteristics, speaker interaction patterns, age-related language indicators, secrecy request and temporal behavior features. More advanced fusion mechanics could also be explored in order to model interactions between semantic and emotional information more effectively.

Finally, the deployed prototype currently analyzes conversations on demand after they have been submitted by a user. Future implementations could investigate real-time monitoring scenarios in which conversations are analyzed incrementally as new messages arrive. Combined with privacy-preserving deployment strategies, local interface mechanisms and minimal data retention policies, such developments could facilitate the creation of practical and ethically responsible online safety systems.

In summary, while the proposed RoBERTa and GoEmotions fusion architecture demonstrates the potential of combining semantic and emotional information for early grooming detection, there remain numerous opportunities to improve generalization, explainability, evaluation methodology and real-world deployment capabilities.

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